

# Worksheet : Logarithms

## Exercise 1

$$\begin{aligned} 1. \log_2 8 + \log_3 27 - \log_7 7 \\ = \log_2 2^3 + \log_3 3^3 - \log_7 7 \\ = 3 \log_2 2 + 3 \log_3 3 - \log_7 7 \\ = 3 + 3 - 1 = 5 \end{aligned}$$

$$\begin{aligned} 2. 2 \log 60 - 2 \log 2 - 2 \log 3 \\ = \log (60)^2 - \log 2^2 - \log 3^2 \\ = \log \left( \frac{60 \times 60}{(4)(9)} \right) \\ = \log \left( \frac{5 \times 60}{9} \right) \\ = \log 100 \\ = \log 10^2 \\ = 2 \log 10 \\ = 2 \end{aligned}$$

$$\begin{aligned} 3. 5 \log_4 2 - \log_4 0.125 - 2 \log_4 8 \\ = \log_4 2^5 - \log_4 \left( \frac{1}{8} \right) - \log_4 (2^3) \\ = \log_4 \left( \frac{2^5}{2^{-3} \cdot 2^6} \right) \\ = \log_4 \frac{2^5}{2^3} \\ = \log_4 2^2 \\ = \log_4 4 \\ = 1 \end{aligned}$$

$$\begin{aligned} ① 4. 2 \log 5 - \log (64)^{\frac{1}{3}} + 4 \log 2 \\ = \log 5^2 - \log (4^3)^{\frac{1}{3}} + \log 2^4 \\ = \log \left( \frac{25 \cdot 16}{4} \right) \\ = \log 100 \\ = \log 10^2 \\ = 2 \log 10 = 2 \end{aligned}$$

## Exercise 2

$$\begin{aligned} 1. \log x + \log (x-2) &= 3 \log 2 \\ \Rightarrow \log x(x-2) &= \log 2^3 \\ \Rightarrow \log (x^2 - 2x) &= \log 8 \\ \Rightarrow x^2 - 2x - 8 &= 0 \\ \Rightarrow (x+2)(x-4) &= 0 \\ \Rightarrow x = -2 \text{ or } x = 4 \\ &\text{not possible} \\ \Rightarrow x &= 4 \end{aligned}$$

$$\begin{aligned} 2. 2 \log_2 5 - \log_2 (x-1) &= 1 \\ \Rightarrow \log_2 5^2 - \log_2 (x-1) &= \log_2 2 \\ \Rightarrow \log_2 \left( \frac{25}{x-1} \right) &= \log_2 2 \\ \Rightarrow \frac{25}{x-1} &= 2 \\ \Rightarrow 2x - 2 &= 25 \\ \Rightarrow 2x &= 27 \\ x &= \frac{27}{2} \end{aligned}$$

$$3. \quad 2 \log_5 x - 3 = -\frac{1}{\log_5 x} \quad (2)$$

$$\Rightarrow 2k - 3 = -\frac{1}{k}$$

$$\Rightarrow 2k^2 - 3k + 1 = 0$$

$$\Rightarrow (2k - 1)(k - 1) = 0$$

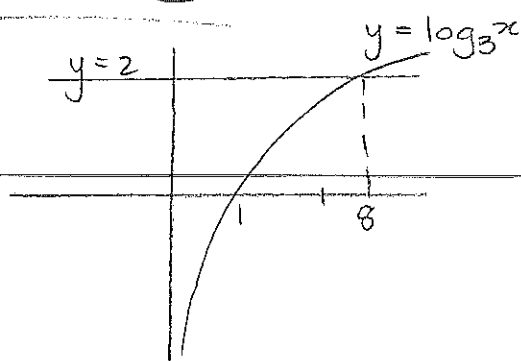
$$\Rightarrow k = \frac{1}{2} \text{ or } k = 1$$

$$\Rightarrow \log_5 x = \frac{1}{2} \text{ or } \log_5 x = 1$$

$$\Rightarrow \log_5 x = \frac{1}{2} \log_5 5 \text{ or } \log_5 x = \log_5 5$$

$$\Rightarrow x = \sqrt{5} \text{ or } x = 5$$

### Exercise 3



Solve  $\log_3 x < 2$ .

First:  $x > 0$  and

$$\log_3 x < 2 \log_3 3$$

$$\Rightarrow \log_3 x < \log_3 3^2$$

$$\Rightarrow \log_3 x < \log_3 9$$

$$\Rightarrow x < 9$$

$$\text{So } x \in (0, 9)$$

### Exercise 4

$$1. \quad \log_2 (3x-2) + \log_2 (x+1) \leq 3$$

First:  $3x-2 > 0$  and  $x+1 > 0$

$$\Rightarrow x > \frac{2}{3} \quad (1) \text{ and } x > -1 \quad (2)$$

Further

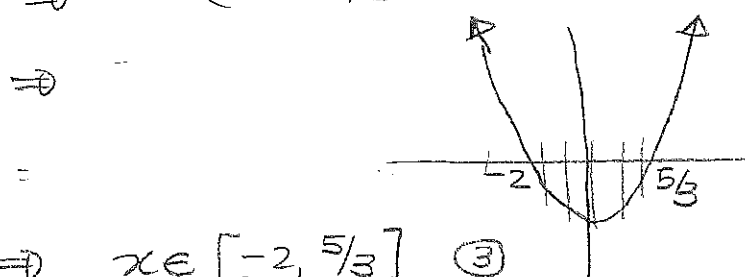
$$\log_2 (3x-2)(x+1) \leq 3 \log_2 2$$

$$\Rightarrow (3x-2)(x+1) \leq 8$$

$$\Rightarrow 3x^2 + x - 2 - 8 \leq 0$$

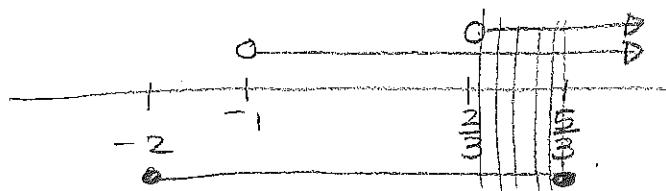
$$\Rightarrow 3x^2 + x - 10 \leq 0$$

$$\Rightarrow (3x-5)(x+2) \leq 0$$



$$\Rightarrow x \in [-2, 5/3] \quad (3)$$

Consider (1), (2) and (3)



$$\text{So } x \in \left(\frac{2}{3}, \frac{5}{3}\right]$$